

LAB 0 – WARMUP [NOT GRADED]

WEEK 1

CONTENT

- Rules and regulations
- Azure Data Studio and AdventureWorks database
- Software installation and testing

NEXT LAB

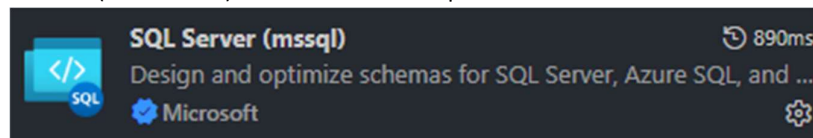
- Transactional data modelling, UML data model – please refresh your knowledge

TUTORIALS

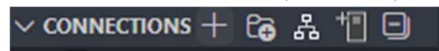
TUTORIAL 1

Download and install Visual Studio Code – it's a quick download and light application; works on all popular operating systems.

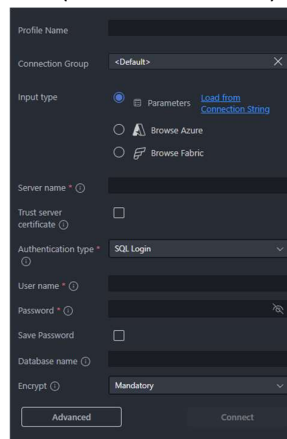
1. Download and install the tool:
 - a. <https://code.visualstudio.com/>
2. Within VS Code:
 - a. Go to extensions (Ctrl+Shift+X) and search for “mssql”. Install this extension:



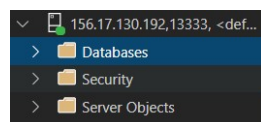
- i.
- b. Open the SQL Server extension tab (Ctrl+Alt+D) and create a new server connection (+):



- i.
- c. Use provided (available on EPortal) connection details (not the ones on the screenshot):



- i.
- d. You should now be able to see databases available on the server. Further we'll focus on AdventureWorks database.



- i.
- e. Play around this interface to get yourself familiar with the type of meta-information that is available here.

- i. For instance, check what kind of indices and keys are defined for the Sales.SalesOrderHeader table
3. Create new SQL query against AdventureWorks database
 - a. Right click on a database and from the context menu select New Query. Double check the name of the currently selected database – combo box on top of the main pane.
 - b. Write a SQL query to get names and list prices of all available products (use table Production.Product, use Name and ListPrice attributes) with list price above 2500USD.
 - c. Run the query – use F5 or Run button.
 - i. Identify the number of returned records and time it took the query to execute.
 - d. Notice that besides the result data grid and server messages, in the result data grid a set of additional options is also available (on the right).



- i.
- e. Try exporting the data into one of the available formats – CSV, XML.
- f. Try creating a simple Bar chart (with vertical data direction)
 - i. If you are seeing an empty chart – check results for proper decimal separator.

TUTORIAL 2 INTRODUCTION TO THE ADVENTURE WORKS DATABASE FOR DATA WAREHOUSING AND DATA DISCOVERY

General overview

Adventure Works database, created by Microsoft, serves as an excellent, real-world example for learning about database systems, particularly in the context of data warehousing and business intelligence. Adventure Works represents a fictional multinational manufacturing company named Adventure Works Cycles. They specialize in the production and sale of bicycles and cycling accessories, operating in global markets.

"The company manufactures and sells metal and composite bicycles to North American, European and Asian commercial markets. While its base operation is located in Bothell, Washington with 290 employees, several regional sales teams are located throughout their market base. In 2000, Adventure Works Cycles bought a small manufacturing plant, Importadores Neptuno, located in Mexico. Importadores Neptuno manufactures several critical subcomponents for the Adventure Works Cycles product line. These subcomponents are shipped to the Bothell location for final product assembly. In 2001, Importadores Neptuno became the sole manufacturer and distributor of the touring bicycle product group. Coming off a successful fiscal year, Adventure Works Cycles is looking to broaden its market share by targeting their sales to their best customers, extending their product availability through an external Web site, and reducing their cost of sales through lower production costs."

Before diving into the technical details, it's crucial to understand the business that Adventure Works simulates. Imagine a company that:

- *Manufactures and sells: A wide range of bicycles (mountain, road, touring) and components.*
- *Operates globally: Has sales and distribution networks spanning multiple countries and regions.*
- *Manages diverse operations: Encompasses product development, manufacturing, sales, marketing, and customer service.*
- *Tracks various aspects of the business: From raw material procurement to final sales and customer interactions.*

This business context is mirrored in the database structure, making it a rich and relevant case study. As such, Adventure Works database contains the scenarios of Manufacturing, Sales, Purchasing, Product Management, Contact Management and Human Resources.

Available flavours

Microsoft provides Adventure Works in two main flavours, each designed for different purposes, which is highly relevant to your data warehousing studies:

- *Adventure Works OLTP (Online Transaction Processing): This version simulates the day-to-day transactional operations of the company. Think of it as the operational database that captures every transaction as it happens. It's designed for:*
 - *Transactional Data Management: Efficiently recording and managing transactions like sales orders, inventory updates, and customer interactions.*
 - *Application Development: Serving as a backend for applications that require transactional data storage and manipulation.*
 - *Relational Database Concepts: Demonstrating best practices in relational database design, normalization, and transactional integrity.*
 - *Data Source: Serving as a source of operational data for further analytical processing.*
- *Adventure Works Data Warehouse (AdventureWorksDW): This is the version specifically designed for analytical purposes and business intelligence. Key characteristics include:*
 - *Analytical Processing: Optimized for complex queries, aggregations, and data analysis over large datasets.*
 - *Business Intelligence and Reporting: Ideal for creating data warehouses, generating reports, and gaining business insights.*
 - *Denormalized or Star Schema Design: Employs a schema optimized for query performance, common in data warehouses, often using star or snowflake schemas.*
 - *Historical Data: Contains a substantial amount of historical data, making it suitable for trend analysis and forecasting.*

AdventureWorks is a sample database created for use in demos and training. As such, it is regularly updated to match the current version of Microsoft SQL Server. Adventure Works 20XX is a database backup of Adventure Works database created in a corresponding SQL Server 20XX version. Please note, that such updates do not affect the structure and content of the data (since AdventureWorks2014 structure and content is the same). Unfortunately, the original samples – as provided by Microsoft – have very limited sales data, i.e., only up to year 2014.

Resources

For basic information about Adventure Works database, please look at the following links:

- *AdventureWorks Sample Databases:*
 - [https://technet.microsoft.com/en-us/library/ms124501\(v=sql.100\).aspx](https://technet.microsoft.com/en-us/library/ms124501(v=sql.100).aspx)
 - <https://msftdbprodsamples.codeplex.com/releases/view/125550>
- *AdventureWorks basic information:*
 - *Business scenarios:*
 - <http://technet.microsoft.com/pl-pl/library/ms124825>
 - *Schemas:*
 - [http://technet.microsoft.com/pl-pl/library/ms124894\(v=sql.100\).aspx](http://technet.microsoft.com/pl-pl/library/ms124894(v=sql.100).aspx)
 - *Data Dictionary:*
 - [https://technet.microsoft.com/en-US/library/ms124438\(v=SQL.100\).aspx](https://technet.microsoft.com/en-US/library/ms124438(v=SQL.100).aspx)
 - <http://www.sqldatadictionary.com/AdventureWorks2014/database.html>
 - https://dataedo.com/samples/html/AdventureWorks/doc/AdventureWorks_2/home.html

We will further utilise Data Dictionary provided by Dataedo. It is a comprehensive document covering all the details of Adventure Works database. This document is available at:

- <https://dataedo.com/download/AdventureWorks.pdf>.

Focus on this document, we will be using it throughout the entire course.

Key Sales Entities

Imagine Adventure Works Cycles not just as a manufacturer, but as a dynamic sales organization. Their primary goal is to sell their bicycles and accessories to customers worldwide. To achieve this, they need to:

- **Manage a Diverse Product Catalog:**
 - Offer a range of bicycles (mountain, road, touring), components, and accessories, each with specific attributes, pricing, and categories.
- **Reach Customers Through Multiple Channels:**
 - Sell directly to consumers online, through reseller partners, and potentially through physical retail locations.
- **Operate in Global Sales Territories:**
 - Manage sales operations across different countries and regions, each with unique market characteristics and customer preferences.
- **Track Sales Performance and Customer Behavior:**
 - Monitor sales transactions, analyze customer purchasing patterns, and understand what drives sales success.
- **Manage Customer Relationships:**
 - Maintain information about customers, their demographics, and interactions to personalize marketing and improve customer satisfaction.

While the Adventure Works OLTP database encompasses various business functions, we'll concentrate on its sales-oriented components. This will allow to narrow down the scope of data discovery process and focus on how this transactional database is structured to handle sales processes effectively. This database is strictly focused on handling everyday operations in the company, e.g., introducing new orders, managing product's inventory, managing customer data, managing sales representatives. As such, this database is geared towards a particular application, or a set of applications, which are used by the employees of the Adventure Works Cycles company. Typically, such applications have dedicated set of views that allow managing all every day-to-day operation, like adding new product, updating product's inventory level, adding new product to an order, adding new order, adding new customer, etc. This is typically done by a set of precooked and hard coded SQL queries within applications. Such queries are highly repetitive, with a well-known and defined structure and affect only a small portion of data – typically only a single row in a table. They are quick and easy to execute, and the relational database system is a perfect fit for such a workload. In short, imagine the database supporting these key sales operations:

- **Taking Customer Orders:**
 - Recording new orders from customers, whether online or through other channels.
- **Managing Product Information:**
 - Maintaining up-to-date details about products, pricing, and availability for sales staff and systems.
- **Processing Sales Transactions:**
 - Handling the details of each sale, including quantities, discounts, shipping, and payment information.
- **Updating Inventory:**
 - Reflecting changes in product stock levels as sales occur.
- **Managing Customer Data:**
 - Storing and updating customer information, addresses, and contact details.

Key Tables in Adventure Works OLTP

Now that you have an introduction to the Adventure Works reality, it's time to get your hands dirty and explore the data itself. This exploration is not just about looking at tables; it's about understanding the story the data tells about Adventure Works Cycles and developing your skills in data manipulation and analysis.

Before we begin, ensure you have the right tools and adopt the right mindset. In the context of the former, focus on database management tools, as you'll need a tool to interact with the database. In particular, you could use SQL Server Management Studio (SSMS) or Azure Data Studio. In the context of the latter, focus on curiosity and a

questioning mind (Approach data exploration with curiosity and ask questions like: "What kind of products does Adventure Works sell?", "Who are their customers?", "Where are their sales concentrated?", "Are there any interesting patterns or anomalies?") and be aware of the business context (try to relate your data findings back to the business context of Adventure Works Cycles, think about how the data reflects the company's operations, challenges, and opportunities).

Let's explore the main tables within the Adventure Works OLTP database that are crucial for understanding its sales structure:

- *Product Tables (Product, ProductCategory, ProductSubcategory, ProductModel): The foundation of sales is, of course, the products being sold. These tables organize product information in a normalized manner:*
 - *ProductCategory: Defines broad product categories (e.g., Bikes, Components, Clothing).*
 - *ProductSubcategory: Provides more specific subcategories within categories (e.g., Mountain Bikes, Road Bikes within Bikes).*
 - *ProductModel: Describes product models, which can be associated with multiple products.*
 - *Product: The central table for individual product listings. It links to categories, subcategories, and models, and contains details like:*
 - *ProductID: Unique identifier for each product.*
 - *Name: Product name.*
 - *ProductNumber: Unique product code.*
 - *StandardCost, ListPrice: Pricing information.*
 - *ProductSubcategoryID, ProductModelID: Foreign keys linking to related tables.*
- *Sales Order Tables (SalesOrderHeader, SalesOrderDetail). Basic sales results from a completed order. The database contains detailed information about orders, i.e., specifying the customer, products, date of order, etc. What is important is that there are two basic types of orders:*
 - *Online. Handled by the company's online system.*
 - *InStore. Order is placed by one of the sales representatives.*

Moreover, each order goes through a series of states before sales completes. There are 6 current statuses possible: 1 = In process; 2 = Approved; 3 = Backordered; 4 = Rejected; 5 = Shipped; 6 = Cancelled. Furthermore, each order can be associated with a certain sales location and sales representative, and each order must contain information about customer's billing and shipment address.

These tables are at the heart of recording sales transactions:

- *SalesOrderHeader: Contains header-level information for each sales order. Think of this as the summary of an order. Key columns include:*
 - *SalesOrderID: Unique identifier for each sales order.*
 - *OrderDate, DueDate, ShipDate: Dates related to the order process.*
 - *CustomerID: Foreign key linking to the customer who placed the order.*
 - *SalesPersonID: Foreign key linking to the employee who handled the sale.*
 - *ShipToAddressID, BillToAddressID: Foreign keys linking to address tables for shipping and billing.*
 - *TotalDue, SubTotal, TaxAmt, Freight: Financial summary of the order.*
- *SalesOrderDetail: Contains line-item details for each product within a sales order. One SalesOrderHeader can have multiple SalesOrderDetail records (one for each product in the order). Key columns include:*

- *SalesOrderID, SalesOrderDetailID: Composite primary key to uniquely identify each line item within an order.*
- *ProductID: Foreign key linking to the product being sold in this line item.*
- *OrderQty: Quantity ordered.*
- *UnitPrice, UnitPriceDiscount: Pricing and discount information for this specific line item.*
- *LineTotal: Calculated total for the line item.*
- *Customer Tables (Customer, Person, Address, CustomerAddress, StateProvince, CountryRegion). As a bicycle manufacturing company, Adventure Works Cycles has two types of customers:*
 - *Individuals. These are consumers who buy products from the Adventure Works Cycles online store. Roughly around 18k customers.*
 - *Stores. These are retail or wholesale stores that buy products for resale from Adventure Works Cycles sales representatives. Roughly around 700 customers.*

Managing customer information is crucial for sales:

- *Person: Contains core information about individuals, which can be customers, employees, or vendors.*
- *Customer: Specifically represents customers. It links to the Person and/or Store table and contains customer-specific details.*
- *Address: Contains address information, used for both customer addresses and company addresses.*
- *CustomerAddress: Links customers to their addresses, allowing a customer to have multiple addresses (e.g., billing, shipping).*
- *StateProvince: Defines states or provinces.*
- *CountryRegion: Defines countries or regions.*
- *Sales Territory (SalesTerritory). Defines sales regions and territories for sales management and analysis. Used to categorize sales geographically and track performance by territory. Contains basic Territory Information: Name, country region code, group, sales YTD, sales last year.*
- *Employee (Employee, SalesPerson). Tracks employees and sales personnel. Defines details for sales orders to track sales representatives for performance and commission calculations.*
 - *Employee: General employee information.*
 - *SalesPerson: Specific details for employees who are salespersons, linking to sales territory and sales quota information.*

Data Discovery and Understanding

Data discovery is the initial, crucial phase in working with any dataset, especially a complex transactional database like Adventure Works OLTP. Data discovery is akin to being a detective – you're exploring an unfamiliar landscape, seeking clues, and piecing together information to understand what data is available, where it resides, and what it represents. Is primarily about breadth and initial reconnaissance. It's about getting a broad overview of the data landscape, identifying key components, and gathering surface-level information. Think of it as creating a map of the territory.

Data discovery is the systematic process of examining and gathering information about a data source. In the context of Adventure Works OLTP, data discovery is about:

- *Inventorying Data Assets: Identifying all the tables, views, and other database objects within the OLTP database, in our case those related to sales.*
- *Profiling Data Characteristics: Understanding the basic properties of the data in each table and column – data types, formats, value ranges, distributions, and the presence of missing values.*

- *Uncovering Data Structure and Relationships: Mapping out the database schema, identifying primary and foreign keys, and understanding how tables are related to each other.*
- *Formulating Initial Hypotheses: Based on your initial findings, start forming preliminary ideas about the data's meaning, potential business rules, and areas for deeper investigation.*
- *Initial Data Quality Assessment: Getting a first impression of the data's quality – are there obvious errors, inconsistencies, or missing data?*

Next step is data understanding, which is not just about seeing the data; it's about making sense of it, interpreting it within the business context of Adventure Works Cycles, and building a comprehensive mental model of how the data works and what it represents. Data understanding is crucial because it forms the foundation for almost every database-related task, from application development to data integration and beyond. It involves:

- *Comprehending Data Semantics: Moving beyond just knowing column names and data types. It's about truly grasping the meaning of each piece of data in business terms. For example, what does a 'SalesOrderID' represent in the real world, what is the significance of a 'ProductSubcategory'?*
- *Discovering Business Rules and Constraints: Uncovering both explicitly defined rules (like database constraints) and implicitly enforced business logic within the data itself. For example, how does Adventure Works ensure data integrity and consistency in its sales processes?*
- *Mapping Data to Business Processes: Connecting the database structure and data content to the actual sales operations of Adventure Works Cycles. For example, how does the data flow reflect the real-world order processing, customer management, and product handling?*
- *Building a Mental Model: Creating a comprehensive and accurate mental representation of the database – its structure, content, relationships, and underlying business logic. This mental model is your key to effectively working with the data.*
- *Assessing Data Quality: Evaluating the fitness of the data for its intended purpose. Is the data accurate, complete, consistent, and valid enough to support reliable sales operations? Are there data quality issues that need to be addressed? We will perform a detailed data profiling process later in the semester, for the time-being we are focusing on understanding the data.*

Let's break down data exploration into structured stages:

- *Stage 1: Schema Immersion - Understanding the Blueprint*
 - *Visual Schema Exploration (ER Diagrams) – Action: Use your database tool to generate or view Entity-Relationship Diagrams (ERDs).*
 - *Identify Table Types: understand the core entity tables (e.g., Product, SalesOrderHeader, Customer).*
 - *Schema Organization: Notice how tables are grouped into schemas. This is a common way to organize database objects logically. Focus on the schemas relevant to sales (Sales, Production, Person).*
 - *Table Names: Familiarize yourself with the table names within the sales-related schemas. Try to mentally link table names to their likely purpose based on your introduction (e.g., SalesOrderHeader likely deals with order headers).*
 - *Column List Preview: Most tools allow you to expand tables and see the column names. Take a quick glance at column names to get a preliminary idea of the data stored in each table.*
 - *Analyze Table Relationships: Trace the lines in the ERD to understand primary key - foreign key relationships. How are tables connected? What are the cardinalities (one-to-many, many-to-many)?*
 - *Data Dictionary Deep Dive – Action: Refer to the data dictionary. For each table and column you encounter in the schema, consult the dictionary for descriptions.*
 - *Column Semantics: Understand the meaning of each column. What business concept does it represent?*

- *Data Types and Constraints:* Note data types (integer, varchar, date, etc.) and constraints (NOT NULL, UNIQUE, etc.). This informs you about the nature of the data and potential data quality rules.
- *Units of Measure:* Pay attention to units for numerical columns (currency, quantities, weights, etc.).
- *Stage 2: Initial Data Glimpse – Getting a Feel for the Data*
 - *Sample Data Retrieval (SELECT TOP/LIMIT) – Action:* Use *SELECT TOP* (SQL Server) or *LIMIT* (other SQL dialects) to fetch a small number of rows from various tables.
 - *Data Value Inspection:* Examine the actual data values. Do they make sense in the business context? Are there any immediate surprises or anomalies?
 - *Data Format Recognition:* Observe how dates, times, currencies, codes, and text are formatted.
 - *Null Value Identification:* Notice columns that contain NULL values. Consider what NULL might mean in each context (e.g., missing information, not applicable).
 - *Exploring Data in Key Columns, Data Types and Range Verification – Action:* Use SQL to verify the data types and ranges of values in key columns.
 - *Understanding Categories:* Identify the different categories, types, or statuses represented in columns.
 - *Value Ranges:* For columns with a limited set of values (like status codes, types), see the full range of possible values.
 - *Data Type Confirmation:* Ensure the data types in the database align with what's described in the data dictionary and what you expect based on the column's meaning.
 - *Range Checks (Numerical Columns):* Find minimum and maximum values for numerical columns to understand their ranges. Are there any unexpected or out-of-range values?
 - *Date Range Analysis (Date Columns):* Determine the earliest and latest dates in date columns to understand the time span of the data.
- *Stage 3: Investigating Relationships and Data Integrity*
 - *Following Foreign Keys (Joining Tables) – Action:* Write JOIN queries to explore the relationships between tables, especially following foreign key constraints.
 - *Relationship Verification:* Confirm that the relationships described in the schema (and data dictionary) are actually reflected in the data.
 - *Data Context Enrichment:* Use joins to combine data from related tables to get a more complete picture. For example, join SalesOrderHeader with Customer to see customer details for each order.
- *Stage 4: Querying and Analysis - Uncovering Insights*
 - *Basic Queries and Filtering – Action:* Write simple SELECT queries with WHERE clauses to filter data based on specific criteria and DISTINCT to see the unique values in categorical columns.
 - *Aggregation and Grouping (Summarization) – Action:* Use aggregate functions (COUNT, SUM, AVG, MIN, MAX) and GROUP BY clauses to summarize data.
 - *Calculate Key Metrics:* Compute totals, averages, counts, etc., for numerical columns. For example, calculate total order values, average order values, total quantities sold, etc.
 - *Group Data by Categories:* Summarize data by grouping on categorical columns to see aggregated metrics per category (e.g., total sales per product category, average order value per customer segment).

- *Joining Tables for Combined Insights – Action: Write JOIN queries to combine data from fact tables and dimension tables to gain richer insights. Identify proper types of JOINS (INNER JOIN, LEFT JOIN) to combine related data effectively.*
- *Stage 5: Inferring Business Rules from Data:*
 - *Based on your data exploration, try to infer business rules or logic that seem to be implemented in the database. Business rule inference is about connecting the technical database structure to the real-world business operations it represents.*
 - *Observe Data Patterns: Look for patterns or constraints in the data that suggest underlying business rules.*
 - *Consider Business Processes: Think about the sales processes at Adventure Works. How are these processes reflected in the data structure and data values?*
 - *Example Inferences:*
 - *"Is it always required to have a customer associated with a sales order?" (Check for NOT NULL constraints and data patterns).*
 - *"Are there specific rules for discounts or pricing based on product categories or customer types?" (Analyze pricing and discount data).*
 - *"What are the typical stages in a sales order lifecycle, based on the 'Status' values and date columns?"*

Now it is time for you to embark on this data discovery journey with the Adventure Works OLTP database. Be systematic, be curious, and be persistent in your quest to unravel the data's story. The deeper your understanding, the more effectively you can leverage the available data for analytics. Please remember that we are focusing solely on sales-related information, as such we can neglect details regarding manufacturing, human resources, purchasing, nor accounting processes.

CURRENT TASKS

TASK 1

We focus solely on the sales process carried out in the Adventure Cycles company. To ease your task, please note that there are two main tables covering information about sale orders: SalesOrderHeader and SalesOrderDetails. Please try performing data discovery of the Adventure Works database, for this purpose use provided Data Dictionary (pdf file) and direct access to the database (query the data!). Use the following questions, as a form of guidance in your discovery process.

1. Using available documentation (PDFs linked above) and having direct access to the database please analyse the AdventureWorks database – take notes on structure and content. Focus on sales/order information – as the main business process we are interested in – and focus on products, clients, employees (salespersons), sales locations – as different entities involved in the main process. Your goal is to answer the following questions:
 - a. Identify main tables and attributes related to orders:
 - i. Where the detailed information about orders is stored?
 - ii. Are there different types of orders?
 1. If yes, what kind of orders are there?
 - iii. Are there different statuses of orders?
 1. If yes, what kind of statuses are there?
 - iv. Which numerical data can be used to measure the *performance* of an order?
 - b. Identify main tables and attributes related to products.
 - i. Which table(s) contain information related to products, like product's name?
 - ii. Are products organised in some manner, like dairy products, clothes, etc.?
 - iii. What additional information about products is available?
 - c. Identify main tables and attributes related to different types of customers.
 - i. Are there different types of customers – name them?
 - ii. How can you identify the type of a customer?
 - iii. How are customers related to sales?
 - iv. What kind of information is available about customers?
 - d. Identify main tables and attributes related to employees handling orders.
 - i. What kinds of employees are associated with sales?
 - ii. How are employees associated with sales?
 - iii. What kind of information is available about employees?
 - e. Identify main tables and attributes related to sales locations.
 - i. Are there different types of sales locations?
 - ii. How are sales locations associated with sales?
 - iii. At what level of detail sales locations are defined – like street, city, country?
 - f. General sales data
 - i. How could you define sales in this database? Where is such sales data located?
 - ii. What is the total sales amount?
 - iii. What years of sales are covered by the data?
 - iv. How many different customers are there?
 - v. How many orders are there?
 - vi. How many products are there?
 - vii. How many employees associated with sales are there?

TASK 2

Using AdventureWorks database please prepare a series of SQL queries:

1. Provide information about the global order amount (money) of the AdventureWorks business.
(Consider using SalesOrderHeader table)
2. Provide information about the global sales amount (money) of the AdventureWorks business.
(Consider using SalesOrderHeader table)

3. Provide information about the yearly sales amount (money) and volume (items sold) of the AdventureWorks business. (Consider using SalesOrderHeader, SalesOrderDetails tables)
4. Provide information about the sales amount in individual years and months in the US region of sales of the AdventureWorks business. (Consider using SalesOrderHeader table)
5. Provide the global profit generated by each salesperson. (Consider using SalesOrderHeader, SalesOrderDetails and Product tables)
 - a. Assume that product's cost (available in Product table) hasn't changed over time.
6. Provide the sales amount generated by each product category, which appeared in more than 10000 orders. (Consider using SalesOrderHeader, SalesOrderDetails and Product tables)